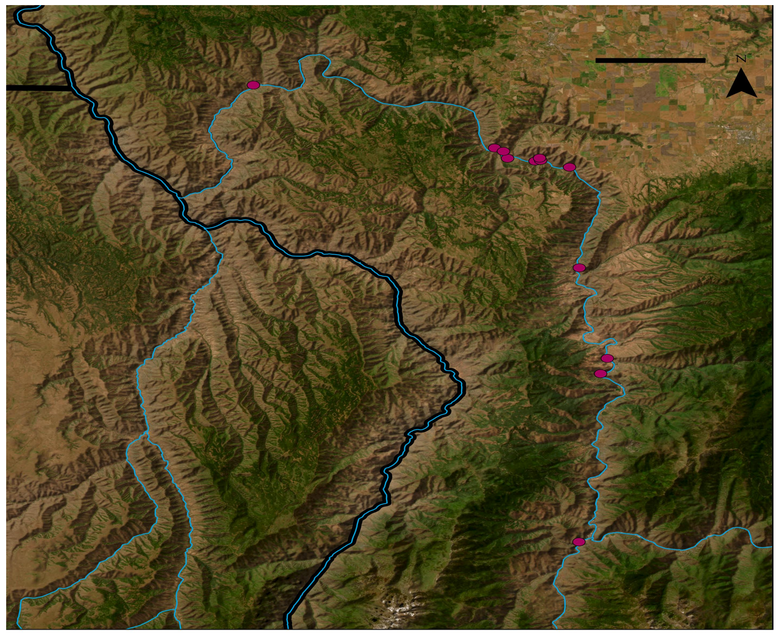
Tree-Ring Growth Response of Netleaf Hackberry (*Celtis reticulata*) to Evapotranspiration and Soil Moisture on the Lower Salmon River, ID

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Introduction

Tree-ring width indices (RWI) are widely used to reconstruct past climate variability, with most studies focusing on precipitation and drought-related variables (Fritts, 1976; Cook and Kairiukstis, 1990). Evapotranspiration (ET) which represents atmospheric water demand has been less explored in dendroclimatology due to its indirect relationship with plant-available water. Netleaf hackberry (*Celtis reticulata*), a drought-tolerant tree found on rocky canyon slopes in the Lower Salmon River Canyon of Idaho, provides a unique opportunity to evaluate growth sensitivity to water availability. Despite its ecological importance *Celtis reticulata* has received very little dendrochronological attention making this one of the first studies of its climate-growth relationships. As climate change is projected to intensify drought across the western United States (Allen et al., 2010), understanding hackberry growth responses becomes increasingly important. This study evaluates the relationship between ET and root zone soil moisture (SMrz) from GLEAM v4.2a (Martens et al., 2017) and hackberry radial growth across 14 sites to assess which variable better predicts interannual growth variability.

Study Area



Methodology

- Increment cores were collected from 169 hackberry (Celtis spp.) trees across 14 sites along the lower Salmon River, Idaho. Two cores per tree were extracted at approximately 20 cm above the ground surface due to the relatively short height of the trees.
- In the laboratory, cores were mounted on wooden supports and sanded with progressively finer sandpaper to clearly reveal annual ring boundaries. Ring widths were then measured using the program CooRecorder. Individual ring-width series were crossdated to ensure accurate year assignment.
- The ring-width series were detrended using a modified negative exponential function to remove biological growth trends related to tree age and size. Standardized Ring Width Index (RWI) chronologies were developed for each site using the dplR package in R.
- Climate data were obtained from the GLEAM v4.2a dataset, which provides global estimates of evapotranspiration (ET) and root-zone soil moisture (SMrz) at a spatial resolution of 0.1° (~10 km). Annual ET and SMrz values (1980–2024) were extracted for each site using geographic coordinates.
- To evaluate climate–growth relationships, Pearson correlation and linear regression analyses were used to test the relationship between RWI and each climate variable (ET and SMrz).

Figure 1 — ET vs Ring Width Index

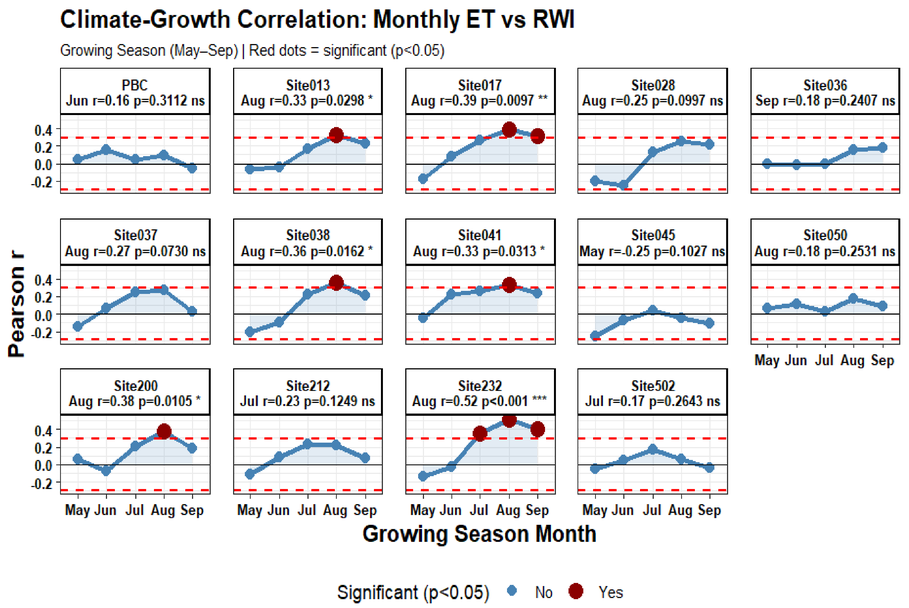
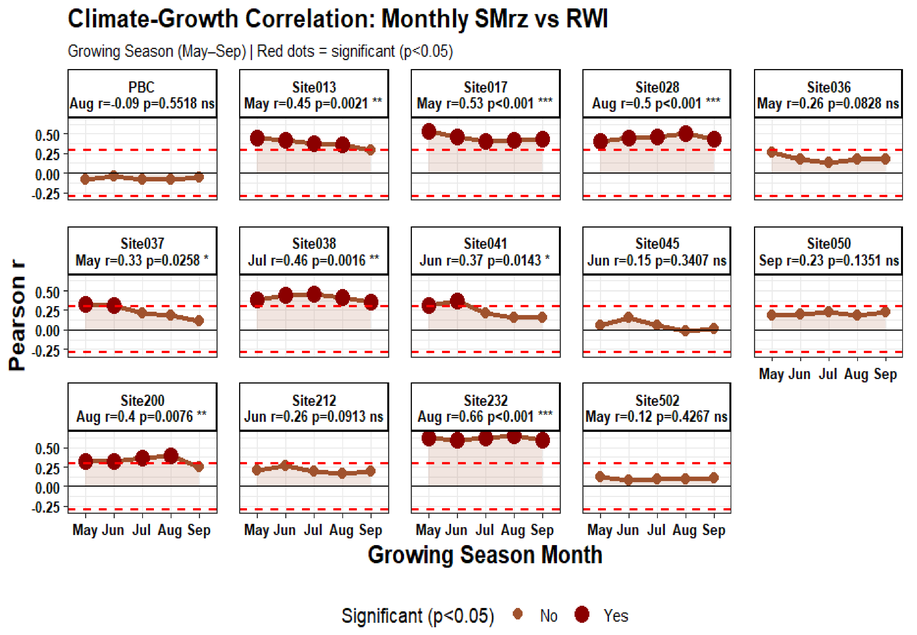


Figure 2 — SMrz vs Ring Width Index



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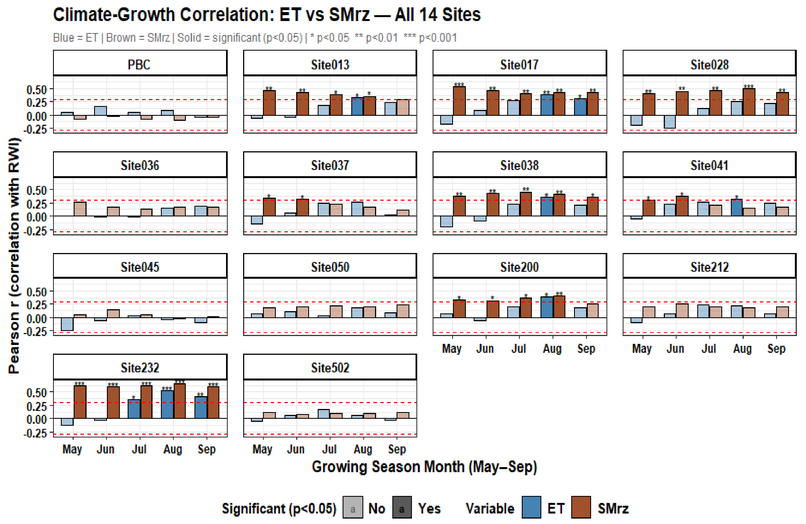
Key Results

ET vs Tree-Ring Growth
- ET significant at 6 of 14 sites
- August = most important month
- Strongest site: Site232 (r = 0.52, p = 0.0003)
- ET explains modest variance — not the primary driver

Soil Moisture vs Tree-Ring Growth
- SMrz significant at 8 of 14 sites
- All significant relationships positive
- Strongest site: Site232 ($R^2 = 0.421$, $p < 0.001$)
- SMrz consistently stronger than ET across all sites

ET vs SMrz Comparison
- SMrz outperformed ET at majority of sites
- SMrz max $R^2 = 0.421$ vs ET max $R^2 = 0.273$
- SMrz is the primary driver of Hackberry growth

Figure 3 — ET vs SMrz Comparison



Conclusion

- SMrz stronger predictor than ET at **9 of 14 sites**
- **August ET** = most critical growth month
- Site232 strongest signal — SMrz R^2 = **0.421**, $p < 0.001$
- ET increasing trend after **2010** = rising atmospheric water demand
- Hackberry growth sensitive to soil moisture variability — **vulnerable to future drought**

Future Work

- **Growing season (JJA) SMrz** analysis using monthly GLEAM data
- Add temperature as additional climate predictors
- Reconstruct past drought using long hackberry chronologies

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